Ek Tasveer — one trusted picture of every child's foundational learning

A NIPUN Bharat decision-support layer built on UDISE+, PARAKH, ASER and state assessments

India @ 71/100 Challenge · Theme 2: Foundational Learning Outcomes · Round 1 proposal — draft v0.3 (4 Oct 2026)

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1. Problem identification

The scorecard. India scores 71/100 on the SDG India Index 2023-24 and 61 on SDG 4 (Quality Education); the adjusted net enrolment rate in elementary education is 96.5 % [1]. NIPUN Bharat (launched 5 July 2021) commits to "universal FLN in primary schools by 2026-27": every child reads with meaning at **≥ 60 words a minute** and handles numbers to 9,999 by the end of Grade 3 [2]. **2026-27 is this school year.**

The lived reality. In rural **government** schools, **23.4 %** of Std III children can read a Std II-level text and **27.6 %** can do at least subtraction (ASER 2024) [3]. Both rose sharply from 2022 (16.3 % and 20.2 %), so the system *can* move — but three in four Grade 3 children in government schools still cannot read a Grade 2 text. In private schools the reading figure is 35.5 % [3].

The measurable gap. *Share of Grade 3 children in government schools who meet the Grade 3 reading and numeracy goals*: mission target 100 % in 2026-27 [2]; independently measured ≈ 23 % (reading) and ≈ 28 % (subtraction) [3].

The underserved group. Children in Grades 1–3 of rural government schools — 68.9 % of India's 14.71 lakh schools are government schools [4] — and, within them, those the data system sees least well: children in **1,04,125 single-teacher schools (33.8 lakh enrolled)** [4], Scheduled Tribe children — the lowest-scoring social group in PARAKH 2024 Grade 3 (maths 57 % vs 61 % for others) [7] — Scheduled Caste children, and children whose home language is not the language of instruction.

The implementation gap — five breaks. Data is not missing: UDISE+ has held student-wise records ("more than 60 fields for each student") since 2022-23 [4]; states run census-style FLN tests; Vidya Samiksha Kendras (VSKs) run dashboards. Yet the chain *recorded → believed → acted on → rechecked* breaks at five points:

Break	Today	Consequence
Commensurability	ASER levels, PARAKH average scores and state tests use different constructs and scales; no system links them (see §2)	Teacher, block and state see different pictures
Trust	Census-style tests are often "proctored and graded by teachers in the same school"; Madhya Pradesh's Pratibha Parv (~6–7 million students) was shown to "mislead inferences about learning levels" [5]	Officials cannot tell progress from reporting pressure
Granularity	Trustworthy surveys (ASER, PARAKH) are samples	Silent about a block, a school, a child
Actionability	Outputs are averages and "% NIPUN"	Nobody is told *which children, which skill,*

Break	Today	Consequence
		which activity, this week
Closed loop	Actions are not logged or rechecked	Nobody learns what worked

2. Evidence and context

Evidence	Finding	Source
ASER 2024 (rural, household, one-on-one oral test)	Std III govt: reading Std II text 20.9 % (2018) → 16.3 % (2022) → 23.4 % (2024); subtraction 20.2 % (2022) → 27.6 % (2024)	[3]
Foundational Learning Study 2022 (NCERT; ~86,000 Grade 3 students, 10,000 schools, 20 languages)	Numeracy: 37 % limited, 42 % sufficient, 10 % superior skills	[6]
PARAKH Rashtriya Sarvekshan 2024 (21,15,022 children, 74,229 schools, 781 districts; Grade 3: 5,99,026 students)	Grade 3 average score, language / maths: rural 64 % / 60 %; **ST children lowest at 61 % / 57 %** vs 66 % / 61 % for others; weakest maths competencies (shapes, money) at 50 %	[7]
Official-data audit, Madhya Pradesh (re-test in 283 schools)	Official census scores inflated by **+38.9 pp (maths)** and **+33.8 pp (Hindi)** over independent re-test levels of 25.1 % and 37.9 %; in Andhra Pradesh, paper tests inflated by 16–20 pp, while tablet-based tests (≈ US$0.50 per test) mostly removed the distortion	[5]
Diagnostic feedback to teachers, Andhra Pradesh (RCT)	Reports on their pupils' results alone: no detectable learning gain	[13]
School-governance reform, Madhya Pradesh (RCT, 1,774 schools)	High compliance with ratings and improvement plans; learning effects null (maths +0.031 SD, Hindi +0.004 SD)	[14]
Teaching at the Right Level (six RCTs, seven states)	Haryana +0.15 SD (language); Uttar Pradesh learning camps +0.6–0.7 SD; scaled to "over 60 million students"	[8], [15]
Phone + SMS targeted instruction, Botswana (RCT)	+0.121 SD numeracy; SMS alone +0.024 SD (not significant)	[16]
Global Education Evidence Advisory Panel (2023)	"Targeting teaching instruction by learning level, not grade" is a *Great Buy*	[17]

Our real-data check (27 states). We linked ASER 2024 and PARAKH 2024 state results. The two national instruments rank states very differently: Spearman ρ = 0.44 for reading/language and 0.26 for arithmetic/maths, with a mean rank gap of ~7 places; Rajasthan is 4th of 27 on PARAKH Grade 3 language (70 %) [22] but 23rd on ASER (12.1 % of government-school Std III children read a Std II text) [3]. Using PARAKH to predict ASER cuts out-of-sample error only from 10.5 to 9.2 points (reading) and 7.9 to 7.6

(maths). The instruments measure different things — so a crosswalk with explicit uncertainty must come before either drives decisions.

Design lesson. Reports and ratings produced compliance, not learning [13], [14]; unverified official numbers were off by over 30 points [5]; diagnosis that becomes a grouped, level-appropriate action works [8], [15], [16]. So the layer must make numbers trustworthy and end every diagnosis in an action.

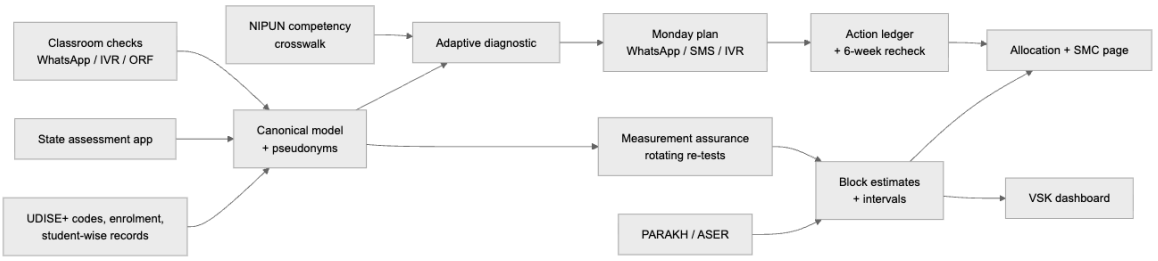
3. Existing ecosystem — why the gap persists

System (deployed)	What it does	What it leaves undone
UDISE+ [4]	School census + student-wise records since 2022-23	No learning levels
VSKs (state command centres)	Command-centre dashboards: attendance, assessments	Monitoring, not a classroom action; no causal evaluation found
State FLN assessments + verification — Haryana *Saksham Ghoshna*: a block is "Saksham" if > 80 % of students score > 50 %, checked by a third-party test in all 119 blocks [10]; Gujarat *Gunotsav*: 3,000 senior officials assess "25% schools from all the blocks" [19]; UP: NIPUN app with external verification [20]	Census data + verification	Verification grades schools or blocks; it is not used to *calibrate* the data, so it must be huge
AI oral reading — Wadhwani AI Vaachan Samiksha, ~30,000 Gujarat schools (developer figures) [11]	Tamper-resistant fluency scores	No independent accuracy check found; no remediation plan
TaRL (Pratham) [8]; chatbots such as SwiftChat in 14 states (vendor figures) [12]	Assess → group → activities; delivery channels to teachers	Needs a light, recurring diagnostic and follow-up at scale

Why the gap persists. Data pipes scaled; verified impact did not. Teacher-graded data is fragile, so states verify by brute force, and verification samples are not used as measurement. Dashboards monitor; few systems turn a diagnosis into next week's activity.

4. Proposed solution — "one picture, two loops"

Ek Tasveer is **not a new app**: a thin decision layer on what states already run (UDISE+, the state assessment app, the VSK, DIKSHA, WhatsApp/SMS/IVR) that adds the five missing links through 12 components with typed, open contracts (JSON Schema) — adoptable one component at a time.



Loop 1 — the classroom, weekly. A teacher runs a few-minute oral check per child on the NIPUN-aligned ladder (letter → word → Grade 1 text → Grade 2 text; numbers → subtraction → division). An adaptive engine stops as soon as it is ≥ 90 % sure of the child's level and returns a **Monday plan**: TaRL groups, one activity per group from the state's own content, and the competency to recheck in four weeks — as one WhatsApp message, ≤ 3 SMS, or an IVR call in the local language. Names stay on the teacher's phone; the system holds only pseudonyms.

Loop 2 — the system, termly ("measurement assurance"). Exam boards already *moderate* school-assessed marks against independent evidence — e.g. NSW's NESA and Queensland's QCAA, which reviews "a selected sample" from every school [21]. We adapt that idea, plus survey back-checks and small-area estimation, to FLN census data, using the sample to *calibrate*, not to grade. Each term a small, rotating sample of schools — chosen by statistical uncertainty and decision value, always with a random component, so every school has a known inclusion probability — is re-tested by the cluster coordinator during the regular academic-support visit, who then coaches that group. A model-assisted estimator calibrates every school's reported results; small-area estimation with UDISE+ covariates then gives **block numbers with intervals that add up to the PARAKH/ASER anchor**. No school-level "trust label" is published: the aim is a true picture, not blame.

Decisions it feeds. CRC weekly visit plans; block/district allocation of remedial camps and training, ranked by expected children helped and weighted for single-teacher schools and SC/ST share; a one-page pictorial summary for each School Management Committee; interval-backed evidence for district Samagra Shiksha plans.

Governance and standards. Keys: UDISE+ school code, LGD codes, a salted pseudonym of the student ID; competency codes from the NIPUN Lakshya Soochi. Role- and purpose-bound access: child-level data only on the teacher's device; cells with < 10 children suppressed; school-level moderation data only for the moderation team. The DPDP Act 2023 requires verifiable parental consent for children's data and bars "tracking or behavioural monitoring of children" unless rules exempt a class of processing (s. 9) [18]; the design holds either way, with pseudonymised, purpose-bound server records and consent captured at enrolment where required. NIPUN asks for "IT based solutions … field level child wise monitoring" built on DIKSHA, UDISE+ and NDEAR standards [2] — this is that layer.

Working demo (real runs; synthetic children — no real child data used). A simulated district (6 blocks, ~600 government primary schools) in which ~30 % of schools inflate teacher-reported results; code, seeds and commands in the repository (`eval/run_eval.py`, `components/*/evaluate.py`).

Result (first runs, 4 Oct 2026)	Value
Adaptive check capped at 8 of 16 items: exact level accuracy (5 levels, both subjects, seeds 101–105)	0.551 — baseline for the auto-research loop

Result (first runs, 4 Oct 2026)	Value
Monday plan for a class of 60, Hindi WhatsApp	659 bytes; SMS version 3 segments; IVR script
Block error of "% meeting goal", teacher-reported vs moderated (10 % of schools re-tested; 5 seeds)	4.09 ± 0.32 pp → 2.34 ± 1.09 pp
Detecting inflating schools (AUC) from public signals only	0.77
Governance policy tests (`components/governance/evaluate.py`)	36 / 36 pass

Honest limits. In a small district (120 schools, 12 re-tested; one seed) moderation *increased* block error (5.4 → 7.7 pp) and intervals under-covered, so the minimum effective sample and calibrated intervals are our next targets; allocation gains are not yet shown. The 5-seed figures used the simulator's default calibration (25 % / 30 %); the ASER-calibrated re-run (23.4 % / 27.6 %) is pending.

5. Feasibility and implementation roadmap

Phase	Scope	What happens	Exit criterion
0 (3 months)	1 district, 2 blocks, clusters staggered	Loop 1 in Grades 1–3; Loop 2 at a 10 % rotating sample; staggered start gives a causal comparison	Teachers use the plan weekly; moderated vs reported gap measured
1 (1 year)	1 state via its VSK	Connectors to the state assessment app and UDISE+; SMC pages; AWP&B evidence	Block estimates with intervals in the state's planning cycle
2 (2–3 years)	Multi-state	Open contracts and components offered as digital public goods	Adopted component-by-component

People and time (design estimates). Teacher: one short oral check per child per term and one message a week. CRC: at a 10 % rotating sample, a block of 100 schools needs 10 school re-tests per term — about 12 children each, inside visits the CRC already makes. Block/district: the existing monthly review uses the moderated numbers. **Costs:** no new devices; SMS/IVR as fallback; tablet-based re-testing was costed at about US$0.50 per test in the Andhra Pradesh study [5]. **Risks → mitigations:** teacher workload → short adaptive checks; language → per-language validation of items and AI scoring; connectivity → offline-first, SMS/IVR; gaming → rotation, random component, no published labels; privacy → pseudonyms, suppression, purpose limits.

6. Expected impact and scalability

Beneficiaries. Grade 1–3 children in government schools of the pilot district, then the state (counted from UDISE+ grade-wise enrolment). **Outcome indicators.** Share of Grade 3 children meeting the reading and numeracy goals at block level, with intervals, every term; teacher minutes per child; share of actions rechecked within six weeks; gap between reported and moderated figures. **Expected effect and scale.** TaRL-style grouping has strong RCT support [8]; the staggered pilot measures our effect rather than assuming it. No new hardware or app: the layer runs on the state's VSK, uses UDISE+ identifiers and speaks WhatsApp/SMS/IVR.

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